

File permissions in Linux

Project description

In this project, I practiced using Linux commands to manage files, directories, software, and file permissions. I used `ls -l` and `ls -la` to inspect file permissions, including permissions for hidden files, and used `chmod` to modify permissions based on the required access. I also used `cd` with absolute and relative paths to navigate between directories and `grep` to search for specific content within files. In addition, I practiced using pipes (`|`) to pass the output of one command to another command, as well as creating, moving, and deleting files and directories. I also used `apt` to install and uninstall software packages.

Check file and directory details

I used `pwd` to display the working directory path and `ls` to display the files and directories in the current directory. I also used `cd` to navigate between directories and `cat` to view the contents of files.

Lab screenshots:

```
analyst@ce8a74719d78:~$ pwd
/home/analyst
analyst@ce8a74719d78:~$ ls
logs  projects  reports  temp
analyst@ce8a74719d78:~$
```

```
analyst@cf4a003f7ec9:~/reports$ cd users
analyst@cf4a003f7ec9:~/reports/users$ ls
Q1_added_users.txt  Q1_deleted_users.txt
analyst@cf4a003f7ec9:~/reports/users$ cat Q1_added_users.txt
employee_id  username  department
1001         bmoreno  Marketing
1026         apatel   Human Resources
1041         cgriffin Sales
1104         mreed   Information Technology
1177         aezra   Human Resources
1188         noshiro Finance
analyst@cf4a003f7ec9:~/reports/users$
```

Describe the permissions string

I learned how to read the permission string displayed by `ls -l`. The permission string shows whether the owner, group, and others have read (r), write (w), or execute (x) permissions. A dash (-) indicates that the corresponding permission is not granted.

Lab screenshot:

```
researcher2@60b1f1e67c3f:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Sep 16 14:07 .
drwxr-xr-x 3 researcher2 research_team 4096 Sep 16 14:23 ..
-rw--w---- 1 researcher2 research_team  46 Sep 16 14:07 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Sep 16 14:07 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Sep 16 14:07 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Sep 16 14:07 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Sep 16 14:07 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Sep 16 14:07 project_t.txt
```

Here, the `project_m.txt` file gives the owner read and write permissions, the group read-only permission, and others no permissions.

Change file permissions

I used `ls -l` to inspect file permissions and used the `chmod` command to modify file permissions as required.

Lab screenshots:

```
researcher2@60b1f1e67c3f:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Sep 16 14:07 .
drwxr-xr-x 3 researcher2 research_team 4096 Sep 16 14:23 ..
-rw--w---- 1 researcher2 research_team  46 Sep 16 14:07 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Sep 16 14:07 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Sep 16 14:07 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Sep 16 14:07 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Sep 16 14:07 project_r.txt
```

```
researcher2@60b1f1e67c3f:~/projects$ chmod g-r project_m.txt
researcher2@60b1f1e67c3f:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Sep 16 14:07 .
drwxr-xr-x 3 researcher2 research_team 4096 Sep 16 14:23 ..
-rw--w---- 1 researcher2 research_team  46 Sep 16 14:07 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Sep 16 14:07 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Sep 16 14:07 project_k.txt
-rw----- 1 researcher2 research_team  46 Sep 16 14:07 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Sep 16 14:07 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Sep 16 14:07 project_t.txt
```

Change file permissions on a hidden file

I used `ls -la` to inspect hidden file permissions and used the `chmod` command to modify file hidden file permissions as required.

Lab screenshot:

```
researcher2@60b1f1e67c3f:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Sep 16 14:07 .
drwxr-xr-x 3 researcher2 research_team 4096 Sep 16 14:23 ..
-rw--w---- 1 researcher2 research_team  46 Sep 16 14:07 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Sep 16 14:07 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Sep 16 14:07 project_k.txt
-rw----- 1 researcher2 research_team  46 Sep 16 14:07 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Sep 16 14:07 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Sep 16 14:07 project_t.txt
researcher2@60b1f1e67c3f:~/projects$ chmod u-w,g+r-w .project_x.txt
researcher2@60b1f1e67c3f:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Sep 16 14:07 .
drwxr-xr-x 3 researcher2 research_team 4096 Sep 16 14:23 ..
-r--r----- 1 researcher2 research_team  46 Sep 16 14:07 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Sep 16 14:07 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Sep 16 14:07 project_k.txt
-rw----- 1 researcher2 research_team  46 Sep 16 14:07 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Sep 16 14:07 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Sep 16 14:07 project_t.txt
researcher2@60b1f1e67c3f:~/projects$
```

Change directory permissions

I used `ls -la` to inspect directory permissions and used `chmod` to modify the permissions using an absolute path.

Lab screenshot:

```
researcher2@60b1f1e67c3f:~/projects$ ls -la /home/researcher2/projects/drafts/
total 8
drwx--x--- 2 researcher2 research_team 4096 Sep 16 14:07 .
drwxr-xr-x 3 researcher2 research_team 4096 Sep 16 14:07 ..
researcher2@60b1f1e67c3f:~/projects$ chmod g-x /home/researcher2/projects/drafts/
researcher2@60b1f1e67c3f:~/projects$ ls -la /home/researcher2/projects/drafts/
total 8
drwx----- 2 researcher2 research_team 4096 Sep 16 14:07 .
drwxr-xr-x 3 researcher2 research_team 4096 Sep 16 14:07 ..
researcher2@60b1f1e67c3f:~/projects$
```

Summary

This project helped me build practical experience with Linux commands for managing files, directories, and permissions. I practiced inspecting and modifying permissions using `ls` and `chmod`, including working with hidden files and directories. I also gained experience navigating the Linux file system, viewing and searching file contents, using pipes, and managing software packages with `apt`.